Status	Finished
Started	Sunday, 25 October 2026, 2:01 PM
Completed	Sunday, 25 October 2026, 2:07 PM
Duration	6 mins 31 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)
b. O(log n)
c. O👎
d. O(1) ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Sequential search
b. Binary search ✓
c. Hash search
d. Linear search

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Checking each element sequentially ✓
b. Dividing the list in half repeatedly
c. Sorting the list before searching
d. Skipping every second element

Question **4**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all

b. When the item is the last element in the array

c. Item is the last element in the array or item is not there at all

d. When the item is somewhere in the middle of the array ✓

Question **5**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Bubble ✓

b. Complexity

c. Insertion

d. Selection

Question **6**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Discovering

b. Mining

c. Finding

d. Searching ✓

Question **7**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is large and unsorted
c. When the list is small and sorted
d. When the list is small and unsorted ✓

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element
b. By comparing each element sequentially
c. By dividing the list length by two ✓
d. By using a hash function

Question **9**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search can find multiple instances of the target element
b. Binary search has a lower time complexity on large, sorted lists ✓
c. Binary search does not require dividing the list
d. Binary search works on unsorted lists

Question **10**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Interpolation search
b. Hash search
c. Linear search ✓
d. Binary search

Question **11**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 500
b. 10 ✓
c. 100
d. 1000

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list contains duplicate elements
b. When the list is very small
c. When the list is unsorted
d. When the list is sorted ✓

Question **13**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list
b. It can be applied only if the list is sorted ✓
c. It compares elements sequentially
d. It works on unsorted lists

Question **14**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Linear
b. Both (1) & (3)
c. Hash
d. Binary ✓

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It requires the list to be sorted ✓
c. It does not work with negative numbers
d. It is slower than linear search for small lists